

CYCLISTIC

Bike-Share Analysis

Report 2026

How do annual members and casual riders use Cyclicistic bikes differently?

639,526

Total Rides Analysed

76.7 %

Annual Members

23.3 %

Casual Riders

13.4 min

Average Ride Length

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Role	Junior Data Analyst
Period covered	January through March 2026
Date	April 2026

Tools used: Excel / WPS Sheets Python SQL (SQLite3) Tableau Public

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Executive Summary

This report examines three months of Cyclistic trip data to understand how annual members and casual riders differ in the way they use the service. The goal is to give the marketing team a clear, evidence-based picture of casual rider behaviour so they can design a conversion strategy that is grounded in what riders actually do rather than what we assume they do.

Between January and March 2026, a total of 639,526 clean rides were analysed using Excel, Python, SQL and Tableau. Eight meaningful differences were identified between the two groups. Three strategic recommendations follow directly from those findings.

Key findings at a glance

- ✧ Members took 76.7% of all rides. Casual riders took the remaining 23.3%, which amounts to nearly 150,000 trips during the quietest season of the year.
- ✧ Casual riders spend an average of 15.6 minutes per trip compared to 11.2 minutes for members, a difference of 38.7%. Longer trips suggest leisure rather than commuting.
- ✧ Members ride most on Tuesdays. Casual riders ride most on Saturdays. These are directly opposite patterns.
- ✧ Casual ridership grew by 255% from January to March, compared to 104% for members.
- ✧ Spring pulls casual riders in far more dramatically.
- ✧ 72.4% of casual rides were on electric bikes, a higher share than members at 66.3%.
- ✧ Members show a clear double commute peak at 8 am and 5 pm. Casual riders show a single midday curve with no morning rush.
- ✧ Navy Pier is the only top-ten station where casual riders outnumber members, confirming that casual use clusters around leisure destinations.

Business Task

Background

Cyclistic is a fictional bike-share company operating in Chicago with a fleet of more than 5,800 bicycles spread across 600 docking stations. Riders can choose between three pricing options: a single-ride pass, a full-day pass, or an annual membership. Those who use the first two options are referred to as casual riders. Those who subscribe annually are referred to as members.

The company's finance team has established that annual members generate considerably more revenue per rider than casual users. Rather than devoting budget to acquiring entirely new customers, the director of marketing, Lily Moreno, believes the more efficient path to growth is converting casual riders who already know and use the product into paying annual members.

Business Task

Analyse Cyclistic trip data to identify the behavioural differences between casual riders and annual members, and use those findings to recommend marketing strategies that will convert casual riders into annual memberships.

Key stakeholders

Lily Moreno, Director of Marketing, is the primary stakeholder and the person who assigned this analysis.

The Cyclistic executive team will review and approve any recommended strategy before it is implemented.

The marketing analytics team will be responsible for executing the campaign once approved.

Question assigned to the analyst

How do annual members and casual riders use Cyclistic bikes differently?

Data Description

Source and licensing

The data was sourced from Motivate International Inc., the company that operates the real Divvy bike-share system in Chicago. It is made publicly available under Motivate's Data License Agreement and represents genuine trip records rather than simulated data. Three monthly CSV files were downloaded and combined for this analysis.

Dataset overview

Detail	Value
Files used	202601, 202602, 202603 (CSV format)
Raw record count	656,274 rides
Clean record count	639,526 rides
Records removed	16,748 (2.5% of the raw total)
Original columns	13
Columns added	5 (ride_length_mins, day_of_week, month, hour, date)
Period covered	January through March 2026
Primary key column	member_casual (values: member, casual)
Data licence	Motivate International Inc. Data Licence Agreement

Credibility assessment (ROCCC)

Before any analysis was carried out, the dataset was evaluated against the ROCCC framework to confirm it was fit for purpose.

Criterion	Assessment	Result
Reliable	Collected directly by Motivate Inc. from live ride systems.	Pass
Original	First-party data with no intermediary processing.	Pass
Comprehensive	Three months only. Winter months underrepresent summer behaviour.	Partial
Current	January to March 2026 is the most recent data available.	Pass
Cited	Publicly licensed and properly attributed.	Pass

Privacy and limitations

Data privacy rules prevent connecting any ride record to an individual rider. This means it is not possible to track how often a single casual rider uses the service, or to determine whether casual riders live within the service area. The seasonal limitation is discussed in Section 07.

Column reference

Column	Description	Type
ride_id	Unique identifier for each trip.	Text
rideable_type	Bike type: classic, electric, or docked.	Text
started_at	Date and time the ride began.	Datetime
ended_at	Date and time the ride ended.	Datetime
start_station_name	Name of the departure station.	Text
end_station_name	Name of the arrival station.	Text
start_lat / lng	GPS coordinates of the departure point.	Float
end_lat / lng	GPS coordinates of the arrival point.	Float
member_casual	Rider type. Either "member" or "casual". Key column.	Text
ride_length_mins	Calculated trip duration in decimal minutes.	Float
day_of_week	Full name of the day the ride started.	Text
month	Full name of the month the ride started.	Text
hour	Hour of day the ride started (0 to 23).	Integer

Methodology

Analytical framework

This analysis followed the six-phase Google Data Analytics process: Ask, Prepare, Process, Analyse, Share, and Act. Each phase produced a documented output that feeds into the next. The full Jupyter notebook, SQL query file, and Tableau workbook are available as companion files.

Tools used

Tool	Purpose
Excel and WPS Sheets	Initial exploration and pivot table analysis to spot early patterns.
Python with Pandas	Combining all three CSV files, cleaning, transforming, and visualising.
Matplotlib and Seaborn	Six publication-quality charts saved for the report.
SQL via SQLite3	Eight structured queries for targeted aggregations and cross-tabulations.
Tableau Public	Two interactive dashboard pages for executive presentation.

Data cleaning log

Every cleaning decision was documented so the process can be fully reproduced. The steps below were applied to the combined three-month dataset.

Step	Action taken	Rows affected
1	Combined three monthly CSV files into a single dataframe.	656,274 total
2	Checked for and removed duplicate ride IDs.	0 duplicates found
3	Converted started_at and ended_at columns to datetime.	All 656,274 rows
4	Calculated ride_length_mins from the two timestamp columns.	All 656,274 rows
5	Removed rides shorter than one minute (false starts).	16,154 removed
6	Removed rides longer than 24 hours (anomalies).	574 removed
7	Extracted day_of_week, month, hour, and date columns.	All remaining rows
8	Standardised member_casual and rideable_type to lowercase.	All remaining rows
9	Removed 20 December outlier rows from the January file.	20 removed
Final	Clean dataset ready for analysis.	639,526 rows

Analysis and Findings

Finding 1

Ride volume split



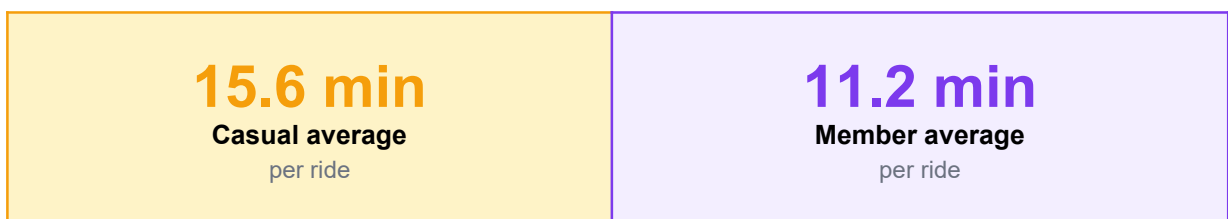
Over the three months studied, members accounted for roughly three in every four rides taken. The casual figure of 148,767 is easy to underestimate, but it represents a substantial pool of engaged riders taking trips during Chicago's coldest season. A casual base this active in winter suggests considerably higher numbers in spring and summer, when leisure cycling peaks.

What this means for marketing

Converting just 10% of current casual riders would add roughly 15,000 new annual memberships without spending a single pound on customer acquisition.

Finding 2

Ride duration reveals intent



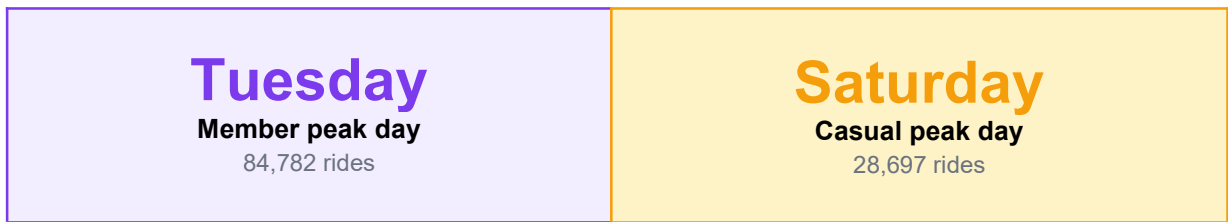
Casual riders take trips that are 38.7% longer than members on average. This single figure tells us a great deal about purpose. Members appear to use the bikes for short, efficient journeys where speed matters more than the experience. Casual riders take their time. Their longer trips are consistent with leisure riding, exploring the city, or cycling to a destination without urgency. The same product is serving two fundamentally different needs.

What this means for marketing

Commuter-focused messaging will not land with casual riders. The pitch needs to speak to the value of longer, more enjoyable rides rather than time savings or efficiency.

Finding 3

Opposite day-of-week patterns



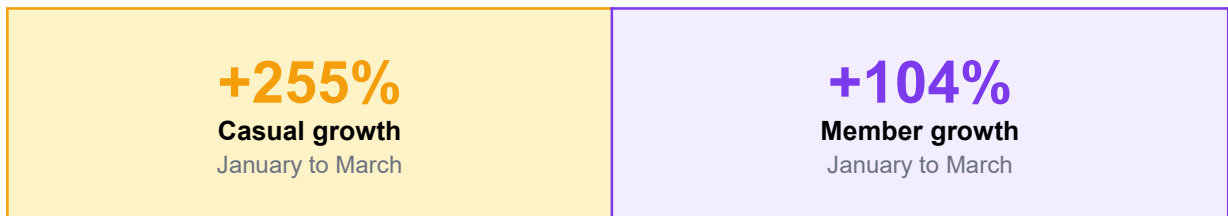
Members ride most in the middle of the working week, with Tuesday, Wednesday, and Thursday being their busiest days. Activity drops noticeably at the weekend. Casual riders show exactly the reverse: quiet mid-week and a clear Saturday peak. On a typical Tuesday, casual riders represent only 17% of all rides. On Saturday that rises to 34%. These two groups are not competing for bikes at the same time because they are using them for entirely different purposes.

What this means for marketing

Weekend activity is the primary window for reaching casual riders. In-app prompts, station promotions, and digital advertising served on Friday evenings and Saturday mornings will reach them at peak engagement.

Finding 4

Casual ridership grows much faster into spring



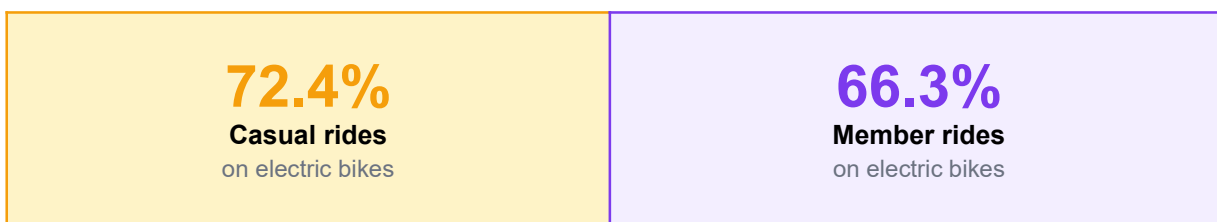
Both groups grew as the weather improved, but casual riders responded far more dramatically. In January, casual riders accounted for under 18% of all rides. By March that had risen to 27.5%. The warmer it gets, the more casual riders appear. Members, by contrast, ride more steadily throughout the year regardless of conditions. This confirms that casual riders are primarily weather-driven leisure users, while members have incorporated cycling into their daily routine.

What this means for marketing

The conversion window opens in spring. A campaign running from March through June reaches casual riders when they are most active, most enthusiastic, and most likely to think about riding more regularly.

Finding 5

Both groups prefer electric bikes, casual riders even more so



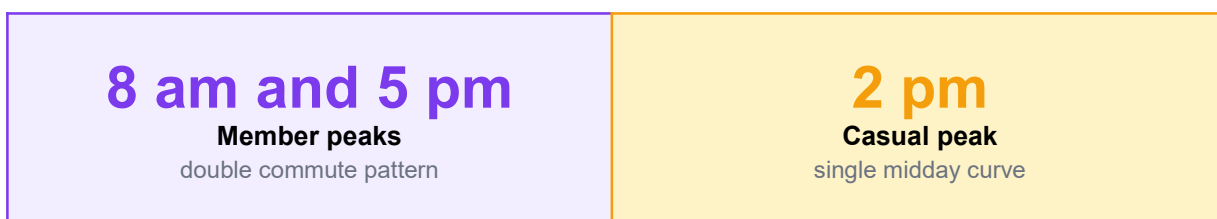
Electric bikes dominate across both groups, accounting for 433,146 of all 639,546 trips. The preference is particularly strong among casual riders, who chose electric bikes for nearly three quarters of their journeys. No docked bikes appeared in the data at all during this period. The lean towards electric among casual riders aligns with their leisure profile: they are choosing the option that requires the least effort and delivers the most comfortable experience.

What this means for marketing

A membership tier with guaranteed electric bike access or a reduced e-bike unlock fee would directly address the strongest revealed preference in casual rider behaviour.

Finding 6

Hour-by-hour patterns tell the full story



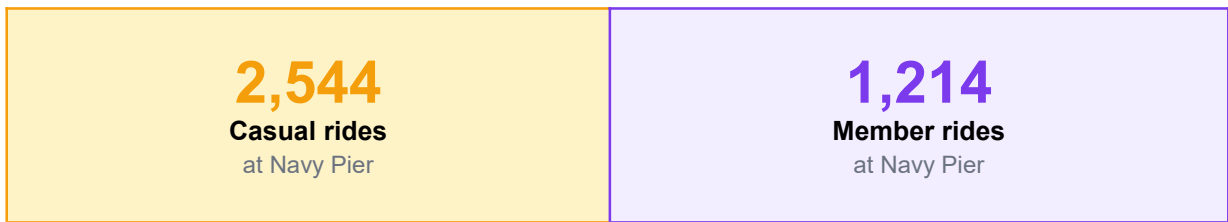
When ride counts are plotted by hour, the difference between the two groups becomes unmistakable. Members show two sharp spikes: one at 8 am with 40,139 rides, and a larger one at 5 pm with 54,226. Between those peaks, numbers fall significantly. This is the pattern of someone cycling to and from work. Casual riders show a smooth, single curve that builds slowly from morning and peaks in the early afternoon around 2 pm to 5 pm. There is no morning rush at all. This is the pattern of someone choosing when to ride based on nothing more than when they feel like it.

What this means for marketing

Promotions served between 11 am and 4 pm on weekends will reach casual riders at the precise moment they are most active on the bikes.

Finding 7

Casual riders cluster around leisure destinations



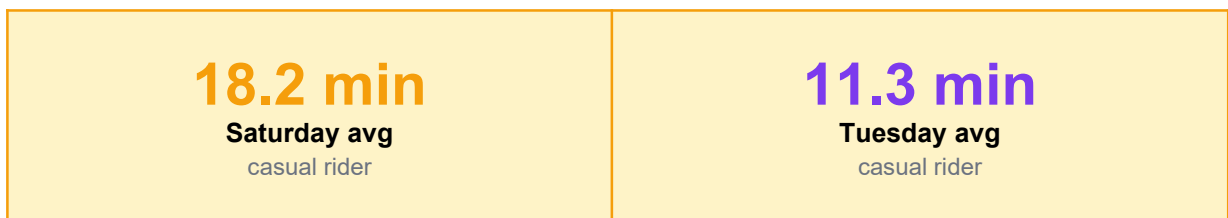
Across the top ten busiest start stations, members outnumber casual riders at every single location except one: Navy Pier, Chicago's most famous waterfront attraction. There, casual riders generated nearly twice the trips of members. The pattern is consistent with everything else in the data. Members tend to start their rides near offices, transport hubs, and residential areas. Casual riders gravitate towards parks, waterfronts, and places people visit for enjoyment.

What this means for marketing

Stations near leisure destinations, particularly Navy Pier, are the highest-value locations for membership sign-up kiosks, promotional materials, and targeted in-app offers.

Finding 8

Casual riders are most engaged at weekends



Casual riders take their longest trips on Saturdays and Sundays, averaging over 18 minutes per ride. During the week their rides shorten noticeably, dropping to around 11 minutes on Tuesdays. Members, by contrast, show much less variation across the week, staying consistently between 10.6 and 12.2 minutes regardless of the day. The weekend surge in casual ride duration suggests that this is when casual riders are most relaxed, most present, and most connected to the experience of cycling itself.

What this means for marketing

Weekend riders are in the right frame of mind to hear a pitch. A message delivered at the end of a long Saturday ride, when the experience is still fresh, is likely to resonate far more than the same message delivered on a quiet Tuesday.

Top Three Recommendations

Each recommendation below follows directly from the findings in Section 05. Together they form a coherent strategy: reach casual riders at the right time, make the financial case in terms that reflect their actual behaviour, and give them a product feature they already value.

Recommendation 1

Run a spring weekend membership drive

- ✧ Casual ridership grew by 255% between January and March and peaks sharply on Saturdays. A targeted campaign running from March through June, focused on weekend riders at high-traffic casual stations like Navy Pier, would reach this audience at the moment they are most active.
- ✧ Offering a limited-time discounted annual membership to riders who have completed three or more rides in the past 30 days targets those who are already habitual enough to convert, rather than wasting spend on one-off visitors.

Expected outcome: the highest conversion rate of any campaign timing, because it aligns with peak casual engagement and improving weather.

Recommendation 2

Build the membership pitch around ride length

- ✧ Casual riders average 15.6 minutes per trip, 38.7% longer than members. This creates a natural, personalised cost argument: a simple in-app message after each ride that shows exactly how much the rider would have saved with a membership, calculated from their actual usage.
- ✧ This approach is more credible than a generic discount offer because it is grounded in the rider's own data. It also reinforces the product's value for the kind of riding casual users actually do.

Expected outcome: a conversion message that feels relevant and specific rather than promotional, which typically yields stronger response rates.

Recommendation 3

Introduce a premium electric bike membership tier

- ✧ 72.4% of casual rides were taken on electric bikes, a higher proportion than for members at 66.3%. A premium membership tier that includes guaranteed e-bike availability or a reduced unlock fee would directly target the feature casual riders already value most.
- ✧ This also gives the business a higher-margin product and creates a clear upgrade path above the standard annual membership, which may attract casual riders who were not yet ready to commit to a basic plan.

Expected outcome: a differentiated offer that speaks to casual riders's strongest revealed preference, and a new revenue stream for the business.

Limitations and Next Steps

Limitations

Seasonal coverage

This analysis covers January through March 2026, which are Chicago's coldest months. Casual ridership is suppressed in winter because leisure cycling falls away in cold weather. The findings almost certainly underestimate the scale of casual activity. A full twelve months of data would give a far more complete picture, particularly of the summer peak when casual riders are likely to dominate far more of total usage.

Individual rider tracking

Privacy rules prevent linking ride records to specific individuals. It is not possible to determine how frequently a given casual rider uses the service, or whether they have used it only once or dozens of times. This limits the ability to segment casual riders by engagement level, which would be valuable for targeting the most convertible users.

Tourist versus resident

The data does not indicate whether a casual rider lives in Chicago or is visiting the city. This distinction matters considerably for marketing channel selection. A tourist is unlikely to buy an annual membership no matter how compelling the offer. Understanding the resident to tourist split among casual riders would sharpen the targeting significantly.

Recommended next steps

- ✧ Expand the analysis to a full twelve months so that summer patterns and year-on-year trends can be examined.
- ✧ Survey a sample of casual riders to understand what has prevented them from buying a membership so far. The data shows how they ride; only they can say why they have not converted.
- ✧ Run an A/B test of the spring weekend campaign at Navy Pier and one or two control stations to measure actual conversion rates before a full rollout.
- ✧ Build a rider segmentation model that flags casual riders who have completed five or more rides in a given month. These are the most engaged users and the most likely to respond to a membership offer.
- ✧ Investigate whether casual riders who exclusively use electric bikes respond differently to the premium tier offer than those who mix bike types.

About this report

This analysis was completed as part of the Google Data Analytics Professional Certificate capstone project. All trip data is publicly available and was provided by Motivate International Inc. under their Data Licence Agreement. Cyclistic is a fictional company used for the purposes of the case study.

Analyst: Savio Kenneth Myers | Date: April 2026 | Tools: Excel, Python, SQL, Tableau